

KONERU LAKSHMAIAH EDUCATION FOUNDATION

Database Systems Engineering and Distributed Backend Development (25CS1302E)

LOST AND FOUND CAMPUS TRACKER

A smart, location-aware recovery network for the campus community

REVIEW 2 • Sec No-10 • Team No-16

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Problem Statement

College campuses generate a constant stream of lost and found belongings — ID cards, electronics, bags, water bottles — but recovery still depends on outdated, fragmented methods.

Result: delayed recovery, lost trust, and repeated losses at the same hotspots across campus.

1

Scattered channels

Notice boards, WhatsApp groups, and word-of-mouth are unorganized and easily missed.

2

No location context

There is no record of where an item was actually lost or found on campus.

3

No verification

Anyone can claim an item — there's no way to confirm true ownership.

4

Slow, manual matching

Students must manually cross-check every posting themselves, item by item.

WHAT THE SYSTEM AIMS TO ACHIEVE

Objectives

1 Build a centralized platform to report and track lost and found items

2 Enable location-aware reporting and hotspot visualization on a campus map

3 Automatically generate potential matches between lost and found reports

4 Provide secure ownership verification before an item is released

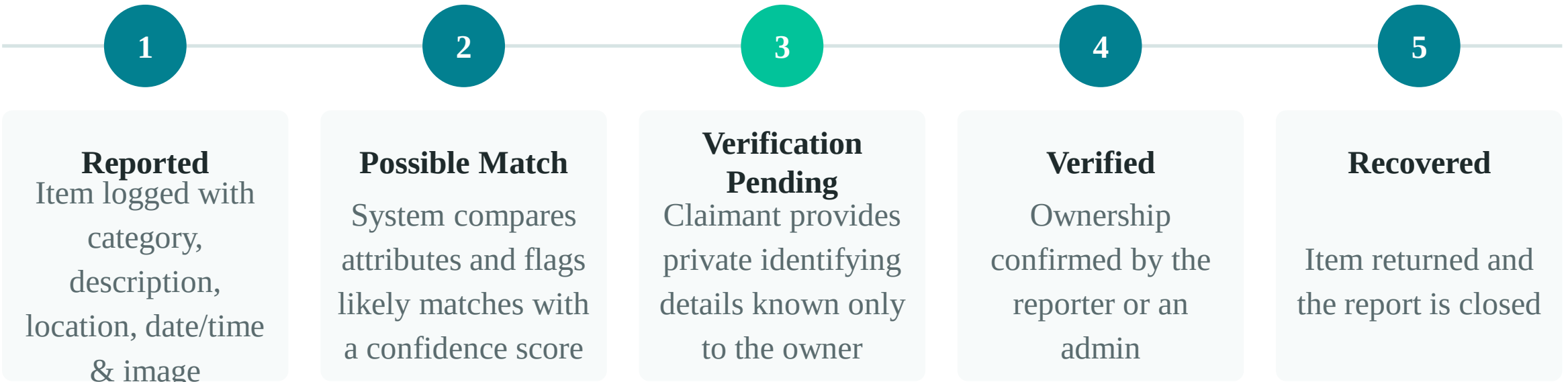
5 Track every report through a clear recovery lifecycle

6 Give administrators tools to review, resolve, and de-duplicate reports

HOW THE PLATFORM WORKS

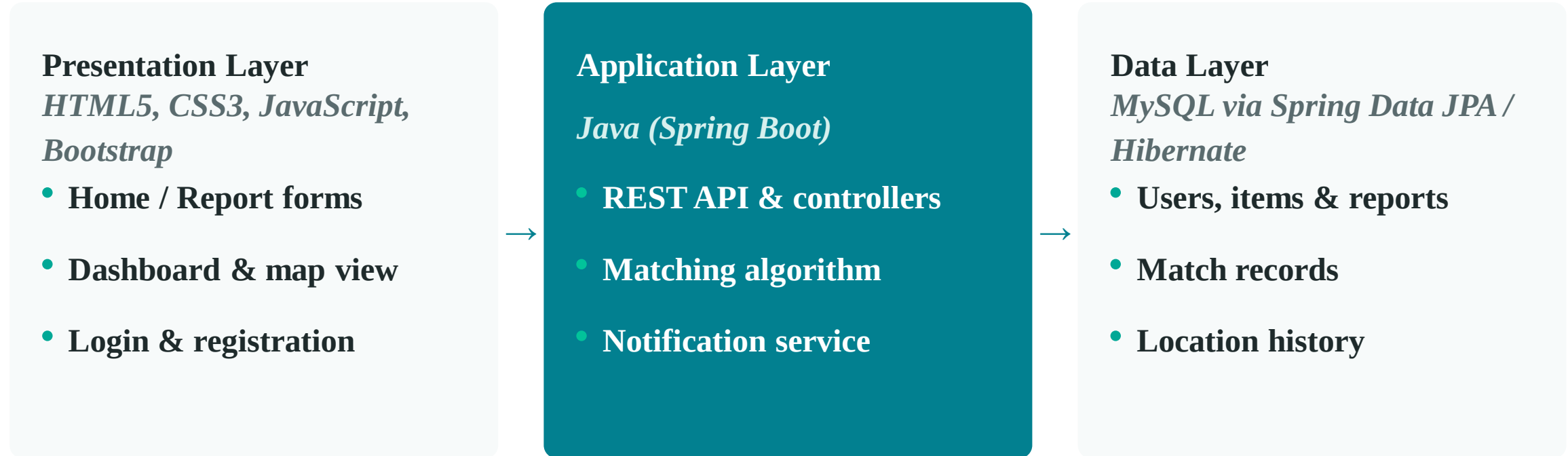
Proposed System

The Lost and Found Campus Tracker gives every reported item a recovery trail — from the moment it's reported to the moment it's safely returned.



THREE-TIER DESIGN

System Architecture



External Services

- Google Maps API — campus location & hotspot mapping
- Email Notification Service — possible-match alerts

CORE BUILDING BLOCKS

Modules / Functionalities

1

User Auth & Profiles

Registration, login and role-based access for students, staff & admins

2

Report Lost / Found Item

Capture category, description, location, date/time and photo

3

Matching Engine

Rule-based comparison of attributes to surface likely matches with a score

4

Location & Map Module

Google Maps-based hotspot view of frequently reported areas

5

Verification & Recovery

Private-detail ownership checks and lifecycle status tracking

6

Notifications

Automated email alerts when a strong match is found

7

Admin Panel

Review flagged reports, resolve disputes, remove duplicates

8

Reporting & Insights

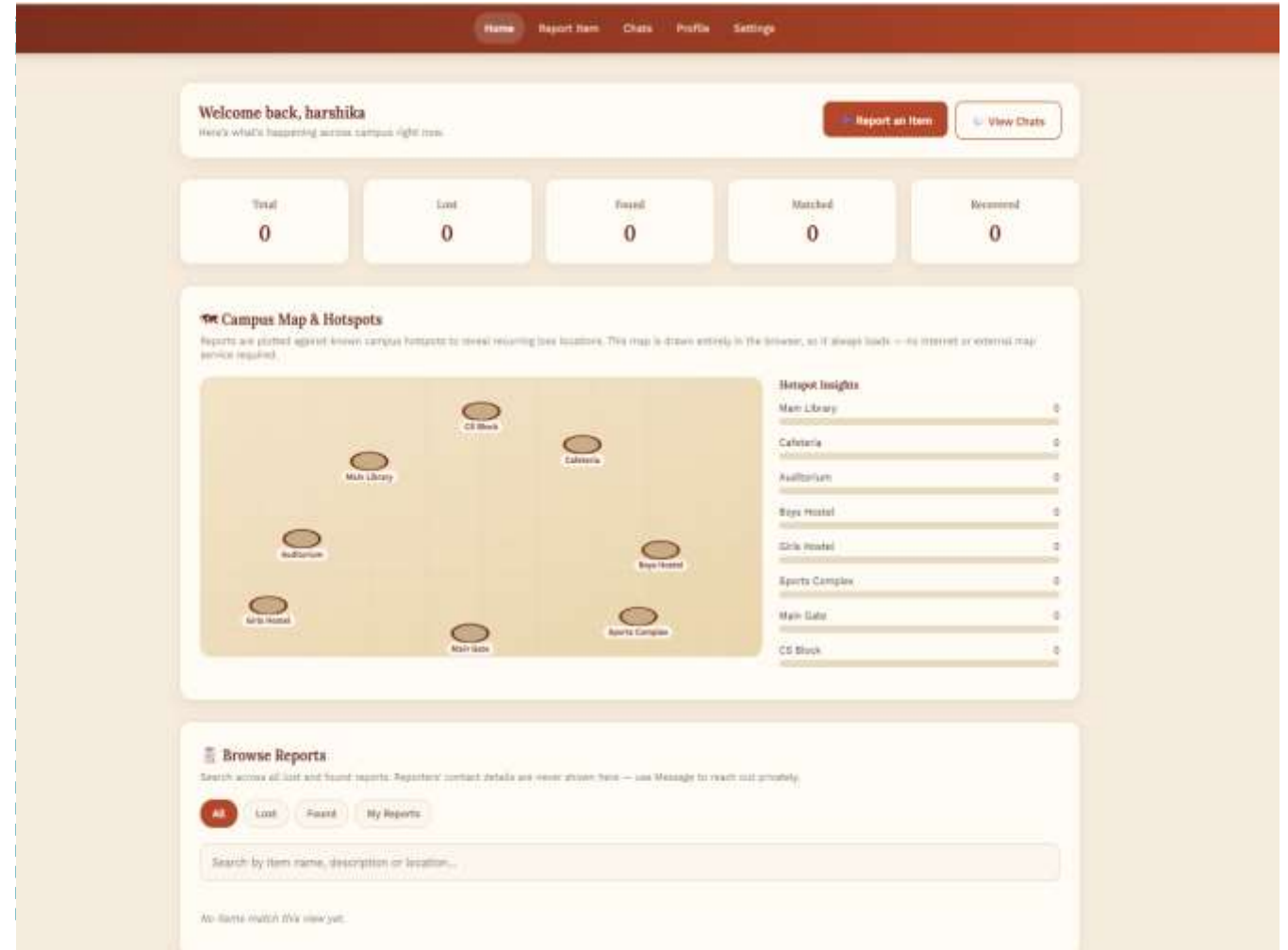
Trends on recurring loss locations and item categories

BOOTSTRAP-BASED UI

Frontend Design — Home Page

Design Highlights

- Clear navigation to Report Lost, Report Found, and Search
- Quick campus-map preview highlighting recent reports
- Recently reported items shown as scannable cards
- Consistent header/footer used across every page

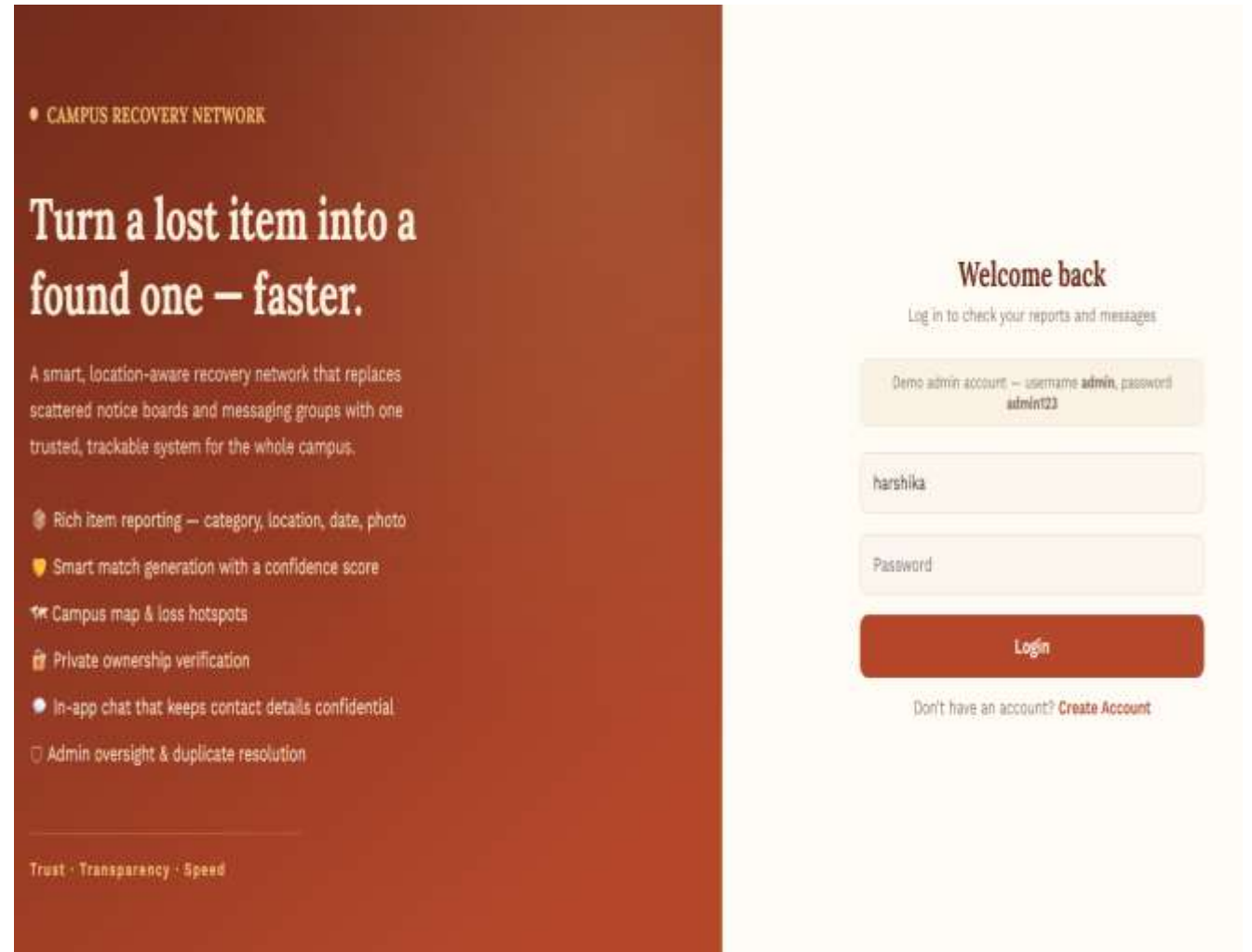


ROLE-BASED ACCESS

Frontend Design — Login / Registration

Design Highlights

- Separate sign-up for students and staff
- Form validation for email, phone and password strength
- Session handling for secure, role-based access
- Clean, distraction-free layout for fast entry

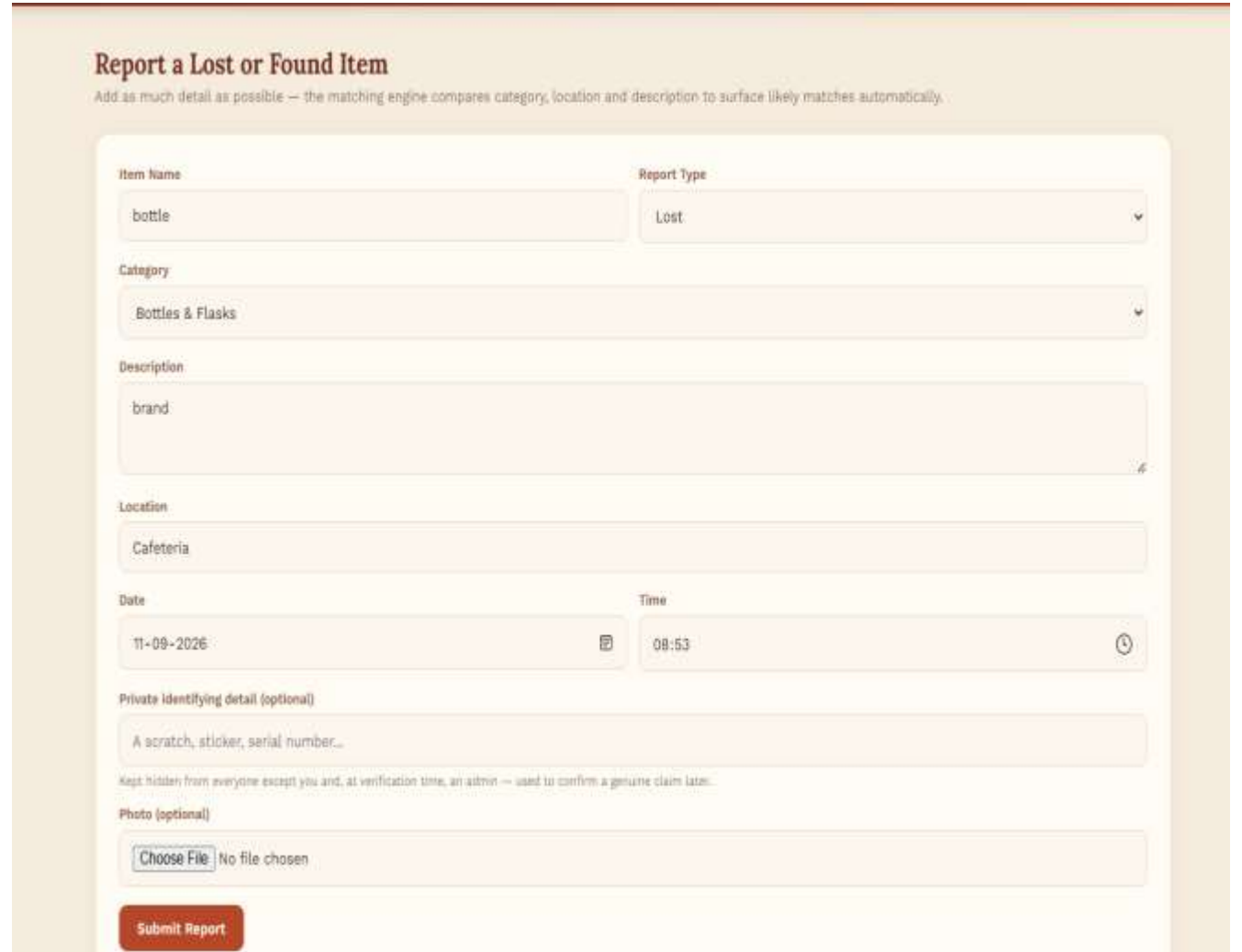


STRUCTURED DATA CAPTURE

Frontend Design — Report Forms

Design Highlights

- Fields for category, description, location, date & time
- Image upload for visual identification
- Optional private detail field for future verification
- Inline validation before submission



Report a Lost or Found Item
Add as much detail as possible — the matching engine compares category, location and description to surface likely matches automatically.

Item Name
bottle

Report Type
Lost

Category
Bottles & Flasks

Description
brand

Location
Cafeteria

Date
11-09-2026

Time
08:53

Private Identifying detail (optional)
A scratch, sticker, serial number...

Kept hidden from everyone except you and, at verification time, an admin — used to confirm a genuine claim later.

Photo (optional)
Choose File No file chosen

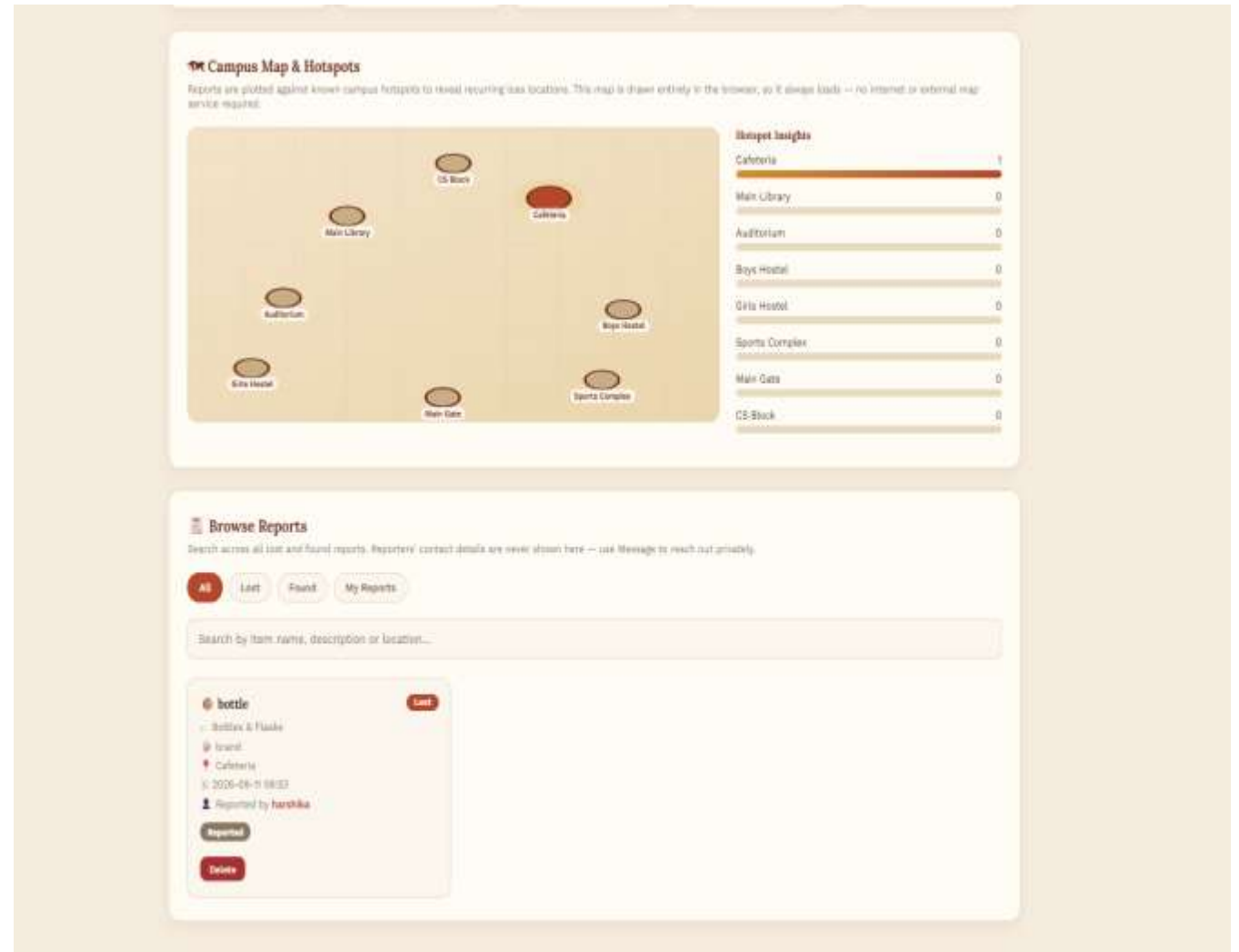
Submit Report

TRACK EVERY REPORT

Frontend Design — Dashboard

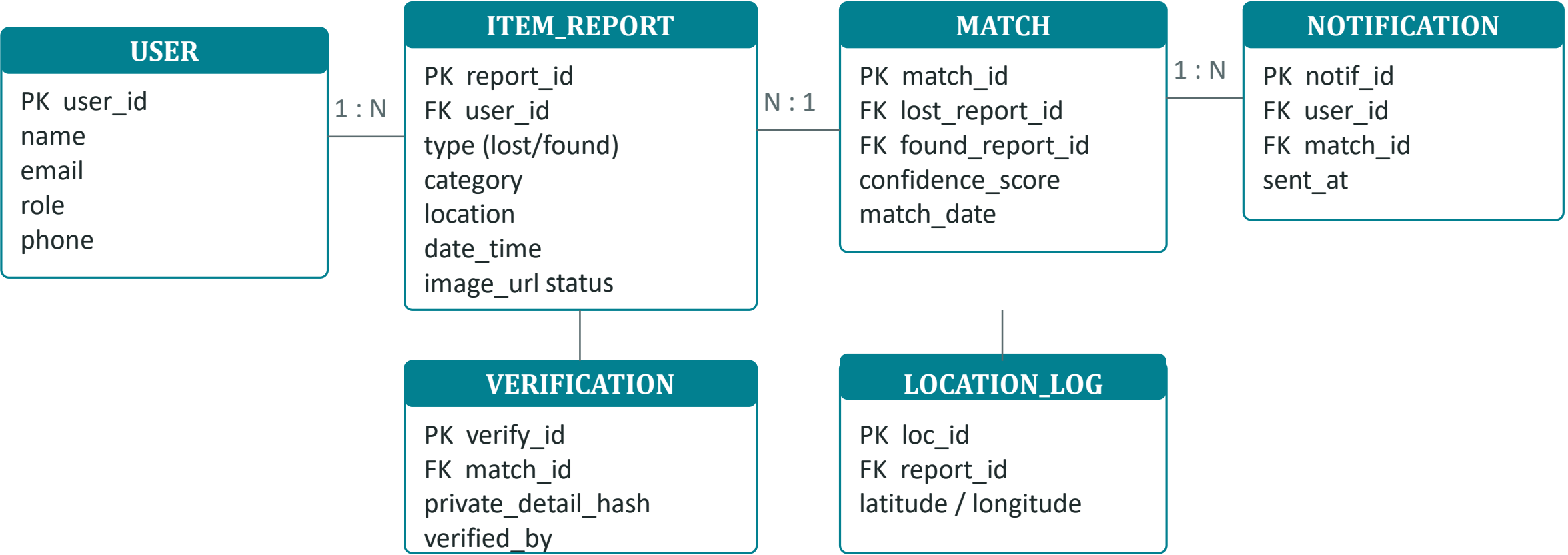
Design Highlights

- My Reports view with live lifecycle status
- Possible-match alerts surfaced at the top
- Filter by category, location or status
- Admin view adds moderation & duplicate controls



CORE ENTITIES & RELATIONSHIPS

Database Design — ER Diagram



MYSQL VIA SPRING DATA JPA

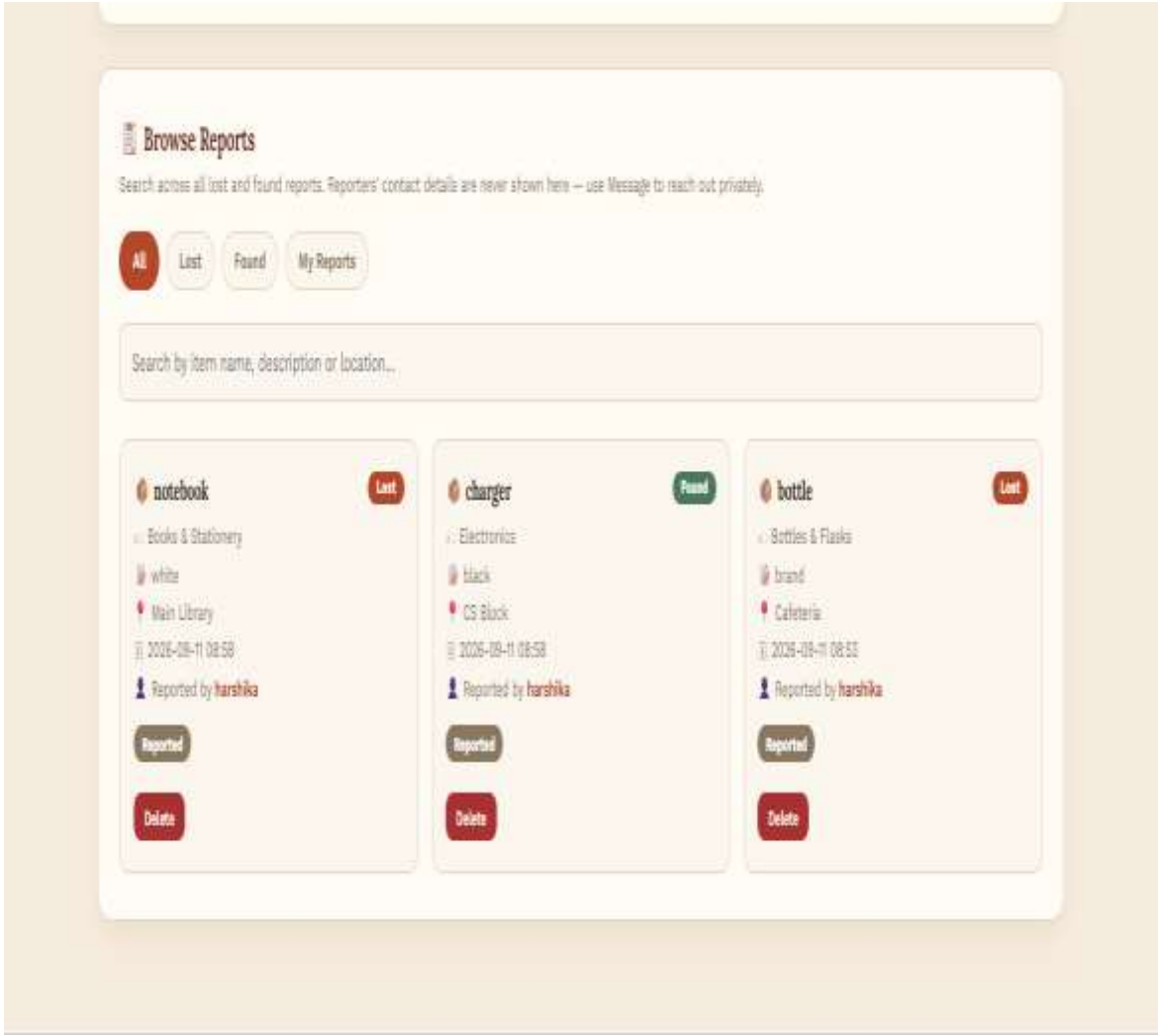
Database Design — Schema & Sample Records

ITEM_REPORT — Key Columns

Column	Type	Notes
report_id	BIGINT (PK)	Auto-increment
user_id	BIGINT (FK)	References USER
type	VARCHAR(10)	'lost' or 'found'
category	VARCHAR(50)	e.g. Electronics, ID Card
location	VARCHAR(120)	Approx. location text
status	VARCHAR(30)	Lifecycle stage

Sample Records (illustrative)

ID	Type	Category	Status
101	Lost	ID Card	Verification Pending
102	Found	Water Bottle	Reported
103	Lost	Earphones	Recovered



STATUS AS OF THIS REVIEW

Current Progress

✓ Completed

- Abstract finalized and submitted
- Requirements, objectives & scope defined
- System architecture designed
- ER diagram & database schema drafted
- Tech stack finalized (Spring Boot, MySQL, Bootstrap)

? In Progress / Pending

- Frontend page implementation (Home, Forms, Dashboard)
- Backend REST APIs & Spring Boot setup
- Matching algorithm implementation
- Google Maps API integration
- Email notification service integration

WHAT'S NEXT

Challenges & Future Work

Challenges / Issues Faced

Designing a matching algorithm that balances accuracy vs. false positives

Keeping ownership verification secure without adding friction

Coordinating schema design across three team members

Future Work

- 1 Complete backend APIs and connect to the MySQL database
- 2 Build and refine all frontend pages with real data
- 3 Integrate Google Maps for live hotspot visualization
- 4 Test the matching engine against realistic sample data

CONCLUSION

Turning campus lost & found into a connected recovery network

- Centralizes lost & found reporting for the entire campus
- Uses location data and rule-based matching to speed up recovery
- Protects ownership through private-detail verification
- Gives administrators visibility, control, and useful insights

Thank You